

ISMS V3 Octopus Cable Connection Key

ISMS Housing Side			Exterior Connection Side					
Pin No.	Wire Base Color	Wire Stripe Color	Routed To	SubConn Connector	Pin Role	Pin No.	Wire Base Color	Wire Stripe Color
1	Black		Vehicle / Topside	MCIL8M	Main Power V+ 1 (VIN1)	1	Black	
2	White		Vehicle / Topside	MCIL8M	Main Power V+ 2 (VIN2)	2	White	
3	Red		Vehicle / Topside	MCIL8M	Main Power GND (VIN-)	3	Red	
4	Green		Vehicle / Topside	MCIL8M	tx/rx Arduino (tx)	4	Green	
5	Orange		Vehicle / Topside	MCIL8M	tx/rx Arduino (rx)	5	Orange	
6	Blue		Vehicle / Topside	MCIL8M	tx/rx RGA (tx)	6	Blue	
7	White	Black	Vehicle / Topside	MCIL8M	tx/rx RGA (rx)	7	White	Black
8	Red	Black	Vehicle / Topside	MCIL8M	Communication GND (shared)	8	Red	Black
9	Green	Black	Fluid Pump	MCIL5F	Fluid Pump ■ GND (Shared)	1	Black	
10	Orange	Black	Fluid Pump	MCIL5F	Fluid Pump ■ Analog (0-5 V)	2	White	
11	Blue	Black	Fluid Pump	MCIL5F	Fluid Pump ■ Power (+12v)	3	Red	
N/a	N/a	N/a	Fluid Pump	MCIL5F	UNUSED Fluid Pump ■ Tachometer	4	Green	
12	Black	White	Fluid Pump	MCIL5F	Fluid Pump ■ Fwd/Rev 0/5v	5	Orange	
13	Red	White	PH Probe	MCIL6F	LX/TX: pH Power V- (GND)	1	Black	
14	Green	White	PH Probe	MCIL6F	LX: USB Data (-) TX: RS485 Data	2	White	
15	Blue	White	PH Probe	MCIL6F	LX/TX: pH Power V+ (+5 VDC)	3	Red	
16	Black	Red	PH Probe	MCIL6F	LX:USB Data (+) TX: pH Analog Signal (0-3.3v)	4	Green	
N/a	N/a	N/a	PH Probe	MCIL6F	UNUSED LX: Shield TX: RS485 Data	5	Orange	
N/a	N/a	N/a	PH Probe	MCIL6F	UNUSED pin for Compatibility	6	Blue	

Compatibility

The 8 pin vehicle/topside connector pinout is backwards compatible with vehicles that connected to prior ISMS versions. The 16 pin main bottle connector, fluid pump, and PH probe connectors ARE NOT backwards compatible.

Locking Sleeves

All Subconn Male (MCILxM) connectors use a female locking sleeve. Conversely the Female Subconns (MCILxF) use a male locking sleeve. This gives the exposed pins a bit of protection.

pH Probe

This cable is default compatible with pyroscience aquapHOx-LX model pH logger but can be adapted to fit aquapHOx-TX model pH sensors w an adapter cable and firmware changes. The pin role for the pH probe is specified based on how the pin would map to the connector on an aquapHOx-LX with "LX:" or aquapHOx-TX with "TX:"